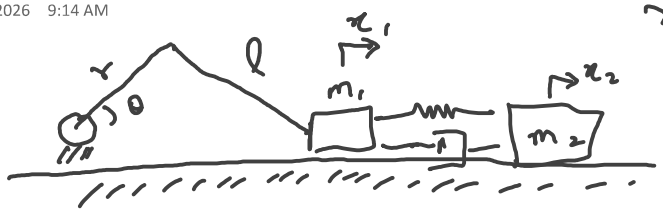
HW1

Friday, January 30, 2026 9:14 AM

Part 1.



$r = 0.15m$

$l = 0.4m$

$\omega = \frac{200 \times 2\pi}{60} = 20.94$ rad/s

1.

$\omega = \frac{d\theta}{dt} =$ constant, x_1 is not a d-o-f, since it is tied to θ.

$$x_1 = r\cos\theta + \sqrt{l^2 - r^2\sin^2\theta}$$

assume at $t=0$, $\theta=0$, therefore $x_1 = r + l = 0.55m$

$$\boxed{\therefore \text{ equation of motion } x_1 = r\cos(\omega t) + \sqrt{l^2 - r^2\sin^2(\omega t)}}$$

(See Fig. 1)

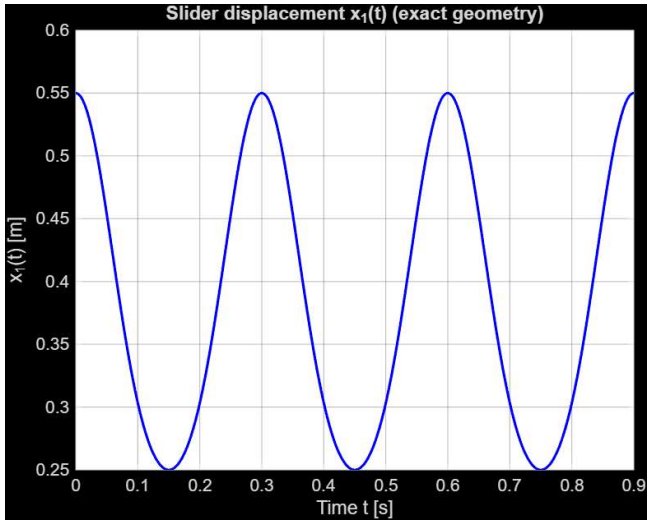


Fig. 1

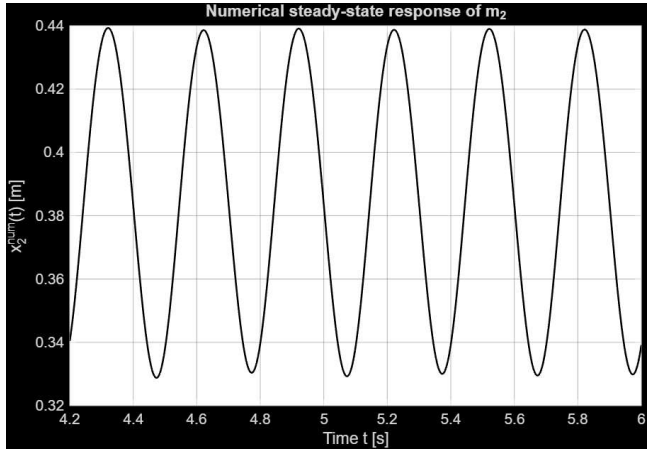
2. $m_2\ddot{x}_2 + c(\dot{x}_2 - \dot{x}_1) + k(x_2 - x_1) = 0$

splitting terms x_1 & x_2

$$m_2\ddot{x}_2 + c\dot{x}_2 + kx_2 = c\dot{x}_1 + kx_1$$

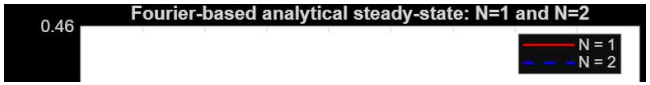
$\therefore m_{eq} = m_2$, $c_{eq} = c$, $k_{eq} = k$

$$f_{eq}(t) = c\dot{x}_1 + kx_1$$



(Fig. 2)

3. Integrate numerically using Matlab ODE45 presenting last 6 cycles in order to remove



the transient effects. (See Fig. 2)

Part 3.

4. $x_1 = r\cos\theta + \sqrt{\ell^2 - r^2\sin^2\theta}$

$= r\cos\theta + L\sqrt{1 - \frac{r^2}{\ell^2}\sin^2\theta}$

set $\frac{r}{\ell} = d$

$\therefore x_1 = r\cos\theta + L\sqrt{1 - d^2\sin^2\theta}$

$\dot{x}_1 = -r\omega\sin\theta - \frac{\ell(-2d\sin\theta\cos\theta)}{\sqrt{1-d^2\sin^2\theta}}\cdot\omega$

$f_{eq} = c\dot{x}_1 + kx_1$

5. set $s = -d^2\sin^2\theta$

$\therefore \sqrt{1+s} = 1 + \frac{1}{2}s - \frac{1}{8}s^2 + \frac{1}{16}s^3 + \dots$ Taylor's expansion

$\simeq 1 - \frac{1}{2}d^2\sin^2\theta - \frac{1}{8}d^4\sin^4\theta$ (using 3 terms)

$\therefore x_1 = r\cos\theta + \ell\left(1 - \frac{1}{2}d^2\sin^2\theta - \frac{1}{8}d^4\sin^4\theta\right)$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$\sin^4\theta = \frac{3}{8} - \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$

$\therefore x_1(t) \approx A_0 + A_1\cos\theta + A_2\cos 2\theta + A_4\cos 4\theta$

where $A_0 = \ell\left(1 - \frac{d^2}{4} - \frac{3d^4}{64}\right)$

$A_1 = r$

$A_2 = \ell\left(\frac{d^2}{4} + \frac{d^4}{16}\right)$

$A_4 = -\ell\left(\frac{d^4}{64}\right)$

Since $f_0(t) = kx_1 + c\dot{x}_1$

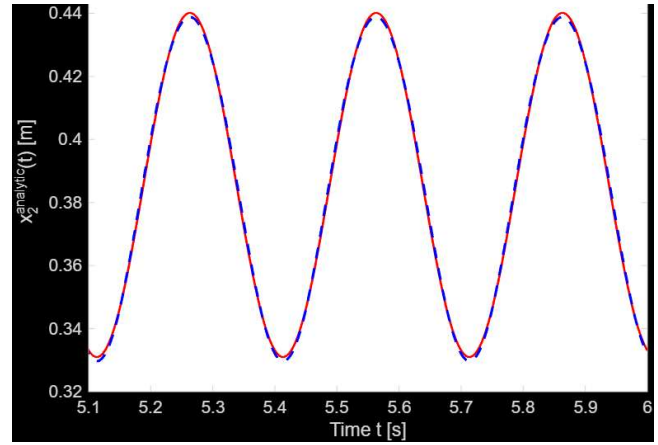


Fig. 3

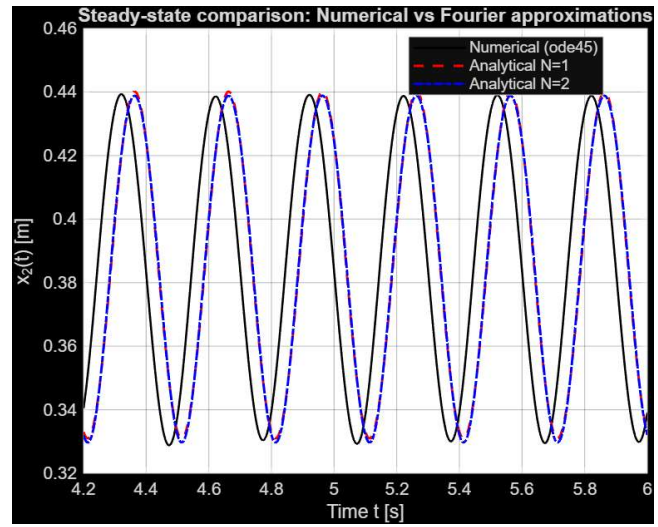


Fig. 4

$$= a_0 + \sum_{j=1}^{4} a_j \cos(j\omega t) + \sum_{j=1}^{4} b_j \sin(j\omega t)$$

where:

$a_0 = kA_0$

$a_1 = kA_1$

$a_2 = kA_2$

$a_4 = kA_4$

$b_1 = -cA_1\omega$

$b_2 = -2cA_2\omega$

$b_4 = -4cA_4\omega$

6. steady state response

x_2 with respect to any frequency j

$$x_{2,j}(t) = \frac{a_j(k - mj^2\omega^2) + b_j j\omega c}{(k - mj^2\omega^2)^2 + (j\omega c)^2}\cos(j\omega t) + \frac{a_j j\omega c - b_j(k - mj^2\omega^2)}{(k - mj^2\omega^2)^2 + (j\omega c)^2}\sin(j\omega t)$$

N=1 approximation

$$x_2^{(1)}(t) = x_{2,0} + x_{2,1}$$

N=2 approximation

$$x_2^{(2)}(t) = x_{2,0} + x_{2,1} + x_{2,2}$$

(plotted as shown in Fig. 3)

7. The ode response & the N=1, N=2 are plotted in Fig. 4

- Taylor's expansion is slightly inaccurate compared to ode45.

- While N_2 captures the peak of the ode45 code correctly, better than N_1